Name: ______________________________

WORKSHEET

Sketching periodic functions – amplitude and period

15	12	10	10	9	15	15

14	15

12	15	12	7	16	16	3

4	2	9

10	12	16	5	14	13	7	4	14	8	13

8	11

10	8	13	4	6	8	16	16	14	13	1

11	7	14	16	12	6	9

Sylvester Stallone

Rule

Given periodic functions $y = A \sin (Bx)$ and $y = \text{A} \cos (Bx)$

A is the **amplitude**

$\frac{2\pi}{B}$ is the **period**

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Questions

1 What is the amplitude of $y = 3\sin(x)$?

2 Sketch this function.

3 What is the period) of $y = \tan(2x)$?

4 Sketch this function.

5 What is the amplitude of $y = 5\cos(x)$?

6 Sketch this function.

7 What is the period of $y = \sin(4x)$?

8 Sketch this function.

For the function $y = 2\cos(3x)$:

9 State the amplitude.

10 State the period.

11 Sketch this function.

For the function $y = 4\tan\left(\frac{1}{2}x\right)$:

12 State the amplitude.

13 State the period.

14 Sketch this function.

15 Sketch the graph of $y = 3\sin(2x)$.

16 Sketch the graph of $y = 7\cos(4x)$.

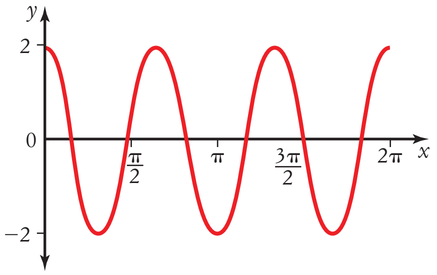
Solutions

A $\frac{\pi}{2}$

C $\frac{2\pi}{3}$

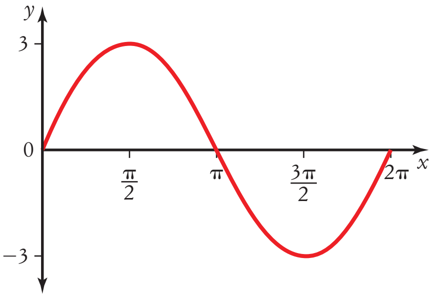
E 2

F



G 3

H



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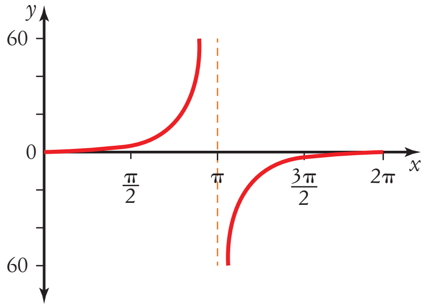

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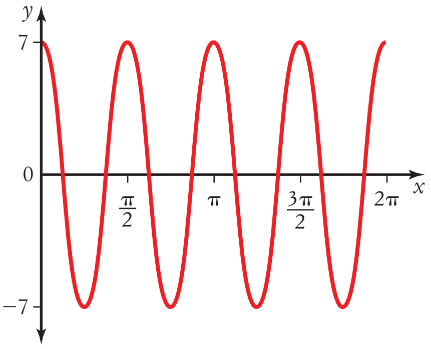
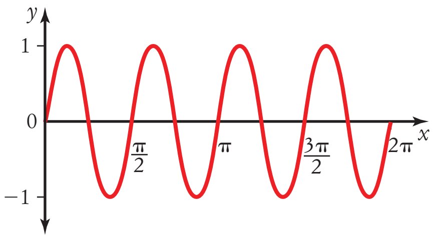


M 5

N 2π

O



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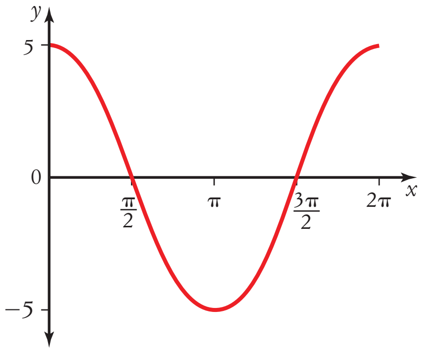

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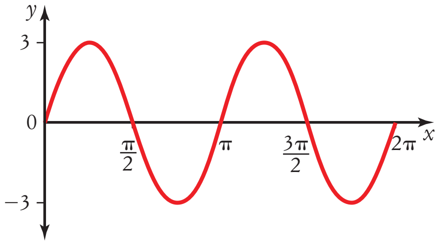

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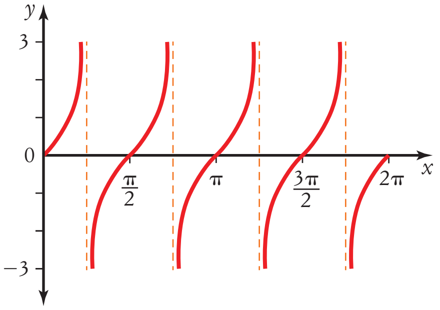

U Stretched vertically by a factor of 4

Y $\frac{\pi}{2}$

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